Complex Adaptive Systems:
a Software Engineering Perspective

Introduction to the Course

Franco Zambonelli, Marco Mamei

What is it About?

- Modern distributed systems are intrinsically "complex"
  - Large-scale, dynamic, situated
- New models and technologies are required
  - Different from traditional distributed systems models (e.g., P2P vs. Client-server)
  - Getting inspirations from natural and social systems (e.g., ants and social networks)
- We will explore this in order to:
  - Understand complexity
  - Attack it!

Course Organization

- Introduction to basic concepts
- From traditional distributed systems to complex adaptive systems (CAS)
- "Lectures" in Complex Adaptive Systems (the rest of the course)

What are "Lectures"?

- Set of introductory seminars focusing on specific topics of CAS
  - Examining basic research results and technologies
  - Supported by selected scientific books/articles
- Lectures imply that:
  - We (the teachers) introduce the basic concepts of a theory/technology during a class
  - You (the students) go over the suggested related books/articles to extract more details

Evaluation Method

- To pass the exam, each student needs to
  - Pass an oral examination covering all the topics and readings of the course
  - Present a short thesis (PowerPoint slides, not a book!!) on a specific topic
    - Theoretical Thesis (e.g., study of a specific aspect via the relevant scientific publications) OR
    - Experimental thesis (e.g., analysis and evaluation of a software tool, set of experiments on a specific phenomena)
- Yes, we are aware this is not simple and harmless…

Detailed Program of the Course

1. Introduction to Complex Adaptive Systems
2. Middleware
3. Context-Awareness and Pervasive Computing Technologies
4. Multiagent Systems
5. Negotiation in Multiagent Systems
6. Tuple-based Coordination

7. Complex Systems
8. Cellular Automata as Complex Systems
9. Complex Networks: Small Worlds Networks
10. Complex Networks: Scale Free Networks

11. Ants and Pheromone-based Coordination
12. Field-based Coordination
13. Routing in Mobile-ad Hoc Networks
14. Sensor Networks
15. The Cloak of Invisibility
16. Computational Self-Assembly and Modular Robots

17. Coordination, Organizations, and Activity Theory (Andrea Omicini, Univ. Bologna)
18. The JADE Multiagent Systems (Federico Bergenti, Università di Parma)
19. Social P2P Mechanisms (David Hales, Università di Bologna)
20. Cellular Automata and Multiagent Systems for Simulation (Stefania Bandini, Università di Milano Bicocca)

Teaching Material

- Slides will be made available on the Web and/or directly mailed to students
- The same for technical and scientific articles
- Web sites with additional material will be signaled whenever possible
- All software will be free
- Hardware freely available for students in our lab

Detailed Program of the Course

1. Introduction: What are CAS and why they are important Franco Zambonelli – 18/1
2. Middleware (TomCat, Jini, CORBA, Sistemi ad Eventi → definizione eventi in Java) Franco Zambonelli 20/1
3. Context-Awareness and Pervasive Computing Technologies (WiFi Bluetooth, RFID, Localization Techniques Trangulation, GPS, Sensori) Marco Mamei 25/1
4. Multiagent Systems (Modelli Generali, Architetture Agenti, Architetture MAS, Sistemi MAS, AOSE) 27/1
5. *Negotiation in Multiagent Systems (Franco Zambonelli)*
6. Tuple-based Coordination (modelli, esempi, sistemi → Jini, JavaSpaces) (Franco Zambonelli) 1/2
7. P2P Systems (Marco Mamei)
8. Sensor Networks (localizzazione, time synch, coverage, power saving)
9. Complex Systems (reaction diffusion)
10. Cellular Automata as Complex Systems
11. Complex Networks: Small World
12. Complex Networks: Scale Free
13. Ants and Pheromone-based Coordination
14. Field-based Coordination
15. Routing in Mobile-ad Hoc Networks
16. The Cloak of Invisibility
17. Computational Self-Assembly and Modular Robots
18. Genetic Algorithms – Learning
19. The JADE Multiagent Systems (Federico Bergenti, Università di Parma)
20. Social P2P Mechanisms (David Hales, Università di Bologna)
21. Cellular Automata and Multiagent Systems for Simulation (Stefania Bandini, Università di Milano Bicocca)